

Data Compression

Learning Objectives

- Understand why data compression is needed
- Distinguish between lossless and lossy compression
- Identify compressed file formats in daily use

Engage (5-7 minutes)

Think of packing clothes in a suitcase: rolling clothes tightly fits more in the same space. Data compression does the same—shrinks files so they take less space and travel faster. Today we learn how this magic works.

Explore (10-12 minutes)

Save a photo as BMP, then as JPEG. Compare file sizes. Open both and look for visual differences. Do the same with a text file saved as TXT versus ZIP. Write which formats reduce size and whether quality changes.

Explain (10-12 minutes)

Compression reduces file size by removing or encoding redundant data. Lossless compression keeps all original data like ZIP files—perfect reconstruction. Lossy compression removes some data permanently like JPEG—slight quality loss but much smaller files. Choose based on whether perfect quality matters.

Elaborate (8-10 minutes)

A school website needs photos. Should they use BMP (large, perfect quality) or JPEG (small, slight loss)? Discuss trade-offs for different uses: medical images, social media photos, text documents. Write recommendations for each scenario.

Evaluate (5-8 minutes)

Exit ticket: Explain the difference between lossless and lossy compression with one example each. Name two compressed file formats.